# Wave speed

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use speed, frequency and wavelength in a consistent relationship.

**SUCCESS CRITERIA - I can...**

1. Use v=fλ with matching units.
2. Check frequency and wavelength changes at fixed speed.
3. Distinguish medium speed from oscillation rate.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “frequency” and “wavelength” to explain your model or result. Sentence starters are optional.

**frequency**: number of cycles each second

**wavelength**: distance between corresponding points on repeated waves

**medium**: the material through which a mechanical wave travels

## 01 Bring back an earlier idea

A system absorbs 70 J heat and does 30 J work.
**Find** its internal-energy change using $\Delta U=Q-W_{by}$.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** v = fλ; λ = v/f.
At fixed wave speed v, increasing frequency f decreases wavelength λ.

**Worked reasoning:** A wave with wavelength 3 m and frequency 2 Hz travels at v=fλ=6 m/s. At the same speed, frequency 3 Hz would require wavelength 2 m.

**Your turn: Draw** both waves over the same 6 m length and label their frequencies.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .

**Marzano check - circle:** ① ② ③

## 03 Look and explain

Two waves travel in the same ideal medium at the same speed. One has twice the frequency.
**Compare** wavelengths.

**Word help:** ideal = using stated simplifying conditions

**Sentence starter:** Both ... . They differ in ... because ... .

**Marzano check - circle:** ① ② ③

## 04 Build the solution

A wave has f=4 Hz andλ=2 m.
**Calculate** v. Another wave has f=8 Hz at that same speed; findλ=v÷f.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Use the idea

A student writes v=f/λ.
**Check** units and explain why that is not a speed.

**Sentence starter:** My claim is … . The diagram/data show …, so … .

**Marzano check - circle:** ① ② ③

## 06 Choose a method

Sunlight carries energy across 4 million km of nearly empty space in a simplified example.
**Identify** the transfer mechanism and explain why matter is unnecessary.

**Sentence starter:** My claim is … . The diagram/data show …, so … .

**Marzano check - circle:** ① ② ③

## 07 Exit check

f=3 Hz,λ=5 m.
**Find** v.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** ① ② ③

## 08 STRETCH: Apply it another way

Create three frequency-wavelength pairs that all give 12 m/s.
**Sketch** one pair.

**Sentence starter:** The … in my model represents … . Its label shows … .

**Marzano check - circle:** ① ② ③